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Instacart 用户行为分析

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Myang

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求职方向：数据分析，数据运营
技能：Python、SQL、Data Visualization

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项目背景

本项目基于 Instacart 的历史交易数据，分析用户在平台上的下单、复购和商品搭配购买行为。项目目标是识别关键消费模式，并将分析结果转化为可执行的业务建议，用于支持推荐策略、补货提醒和促销时机优化。

核心业务问题

- 用户通常在什么时间下单?
- 哪些商品或品类表现出更强的复购特征?
- 哪些商品经常被一起购买?
- 如何基于这些行为特征优化推荐和营销策略?

数据、方法与工具

数据

Instacart 公开交易数据集，包括 orders、products、aisles 等相关表。

方法

数据清洗、探索性数据分析（EDA）、购物篮分析、复购分析、商品共购分析。

工具

Python、pandas、matplotlib、SQL

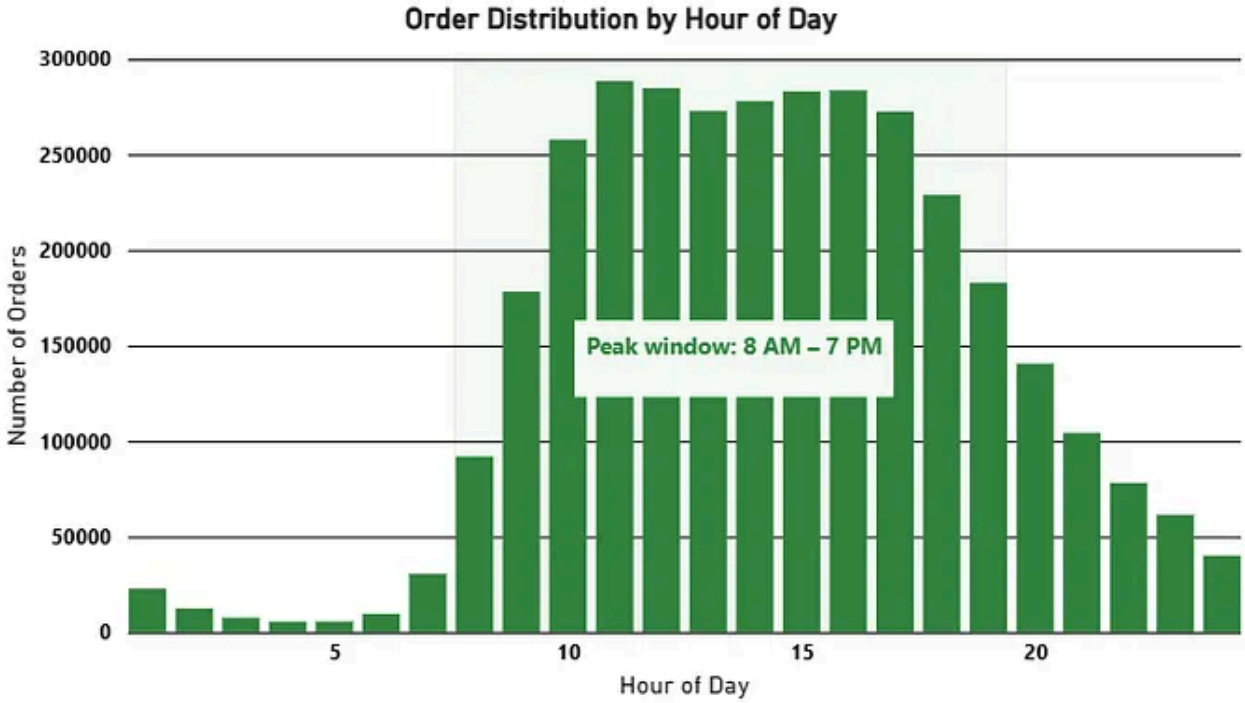
角色

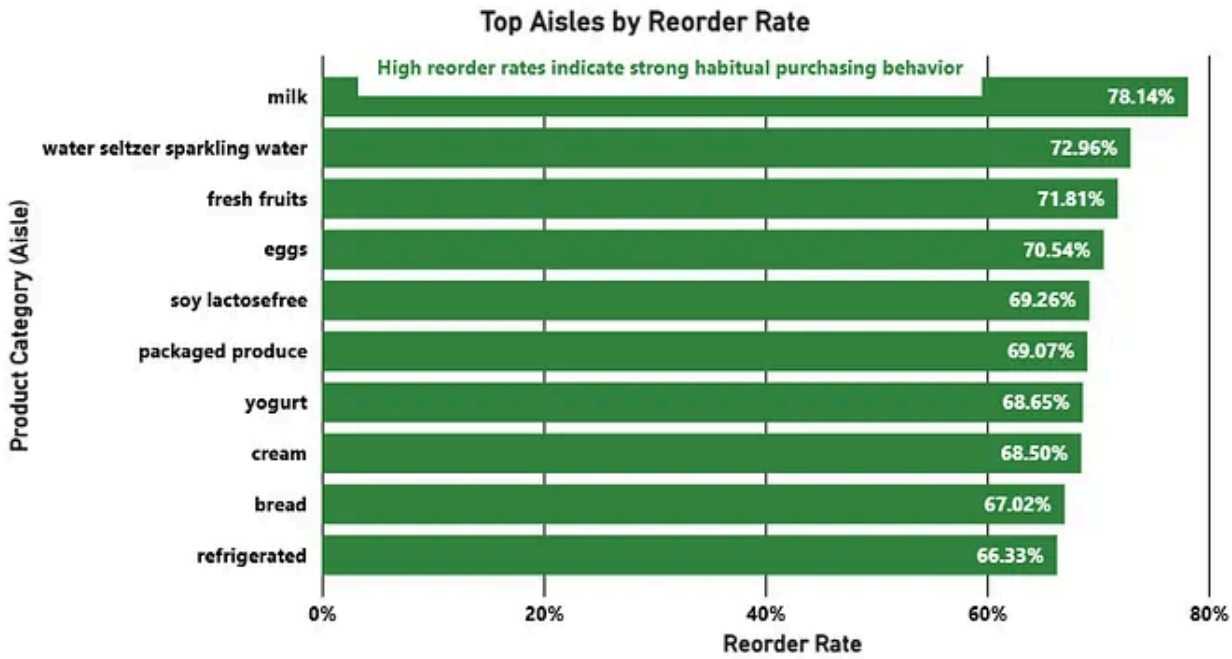
独立完成数据整理、分析、可视化及业务洞察输出。

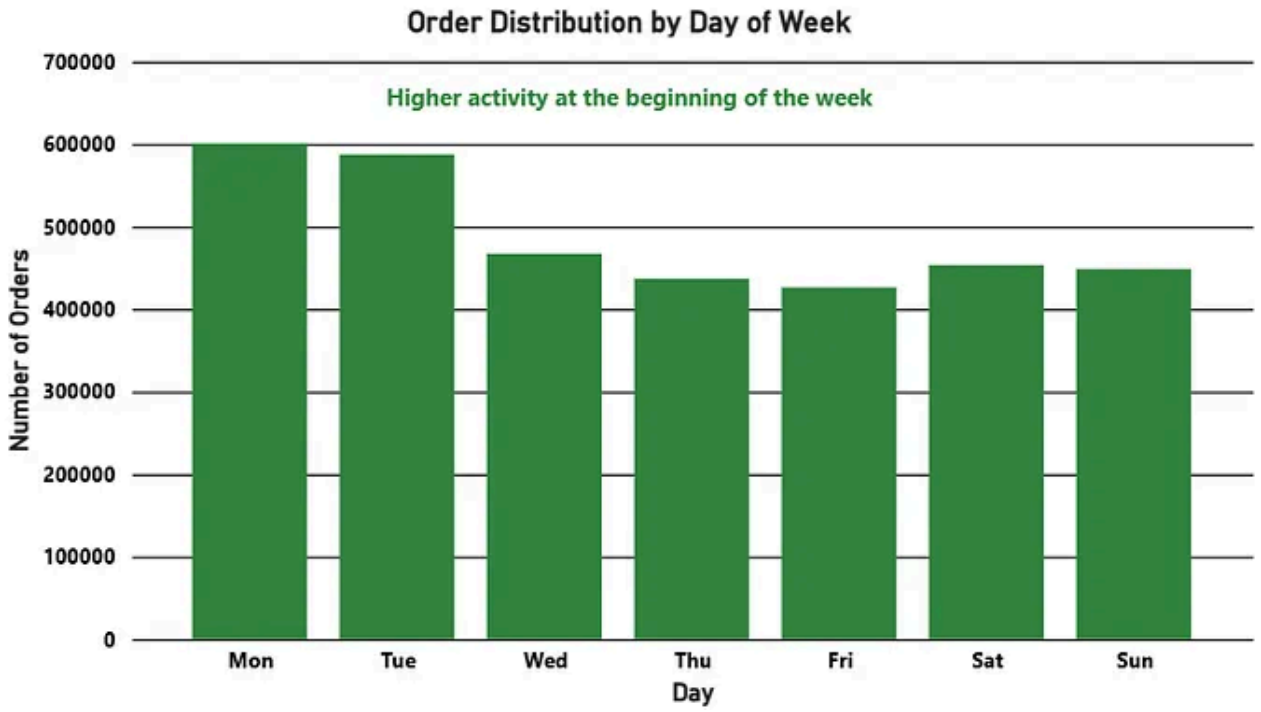
方法说明

首先，我对订单表、订单商品表和商品信息表进行了清洗与整合，统一关键字段并构建可分析的数据集。随后，从三个方向展开分析：

1. **下单行为分析**：观察用户在一天中的下单高峰，以及一周内的下单分布。
2. **购物篮与复购分析**：分析每次订单的商品数量，以及不同商品和品类的复购表现。
3. **共购分析**：识别在同一订单中高频共同出现的商品组合，用于发掘潜在的交叉销售机会。


Order Distribution by Hour of Day
300000
250000
200000
150000
100000
50000
0
Number of Orders
Peak window: 8 AM – 7 PM
5
10
15
20
Hour of Day


Top Aisles by Reorder Rate
High reorder rates indicate strong habitual purchasing behavior
Product Category (Aisle)
milk
78.14%
water seltzer sparkling water
72.96%
fresh fruits
71.81%
eggs
70.54%
soy lactosefree
69.26%
packaged produce
69.07%
yogurt
68.65%
cream
68.50%
bread
67.02%
refrigerated
66.33%
0%
20%
40%
60%
80%
Reorder Rate



分析结果

1. 用户下单时间具有明显集中性

从订单时间分布来看，用户下单主要集中在白天时段，尤其是上午到傍晚之间。这说明平台的营销触达和推荐展示应优先围绕高活跃时段进行优化，以提高曝光和转化效率。

2. 周初活跃度略高，呈现规律性采购特征

从一周内的订单分布来看，用户在周初的下单活跃度略高，说明一部分购物行为与家庭周期性补货相关。这种模式适合用于设计固定节奏的提醒机制，例如按周补货或高频品类推荐。

3. 购物篮规模反映高频、轻计划式购物特征

订单商品数量整体相对集中，表明用户的购物行为更偏向高频、小批量采购，而不是一次性大量囤货。这意味着推荐策略应更加关注“即时需求”和“高相关商品补充”，而不是单纯推更多商品。

4. 部分品类复购率较高，适合做补货提醒

复购分析显示，一些高频消费类商品和品类具有更强的重复购买特征。这类商品更适合作为个性化补货提醒的重点对象，例如在用户可能用完之前进行触达，从而提高复购率和用户黏性。

5. 高频共购组合体现明显的搭配购买模式

共购分析发现，部分商品在同一订单中经常同时出现，说明用户在购物时存在较稳定的搭配关系。这为平台提供了构建 bundle recommendation 和 cross-sell 策略的依

据，有助于提升客单价和推荐相关性。

业务建议

基于以上分析结果，我提出以下三点优化方向：

1. 个性化补货提醒

针对高复购商品和品类，建立基于购买周期的提醒机制，帮助用户在合适的时间完成再次购买。

2. 搭配推荐与组合销售

基于高频共购商品组合，设计相关推荐模块或商品捆绑展示，提升交叉销售效果。

3. 高活跃时段促销触达

在用户下单高峰期加强推荐曝光和促销活动投放，以提高点击率和转化率。

结论

本项目通过对 Instacart 用户交易数据的分析，识别了用户在下单时间、复购行为和商品搭配购买上的关键特征，并将这些特征进一步转化为推荐和营销优化建议。整体来看，用户行为具有较强的规律性，这为平台在推荐系统、补货机制和促销策略上的优化提供了明确方向。

项目链接

GitHub Repository

完整代码、数据处理流程与分析过程已整理至 GitHub。

myang2003820-lang/Instacart-reorder-basket-analysis: Customer reorder behavior and market basket analysis using Instacart transactional data

数据分析

Y

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Written by Myang

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Y Myang

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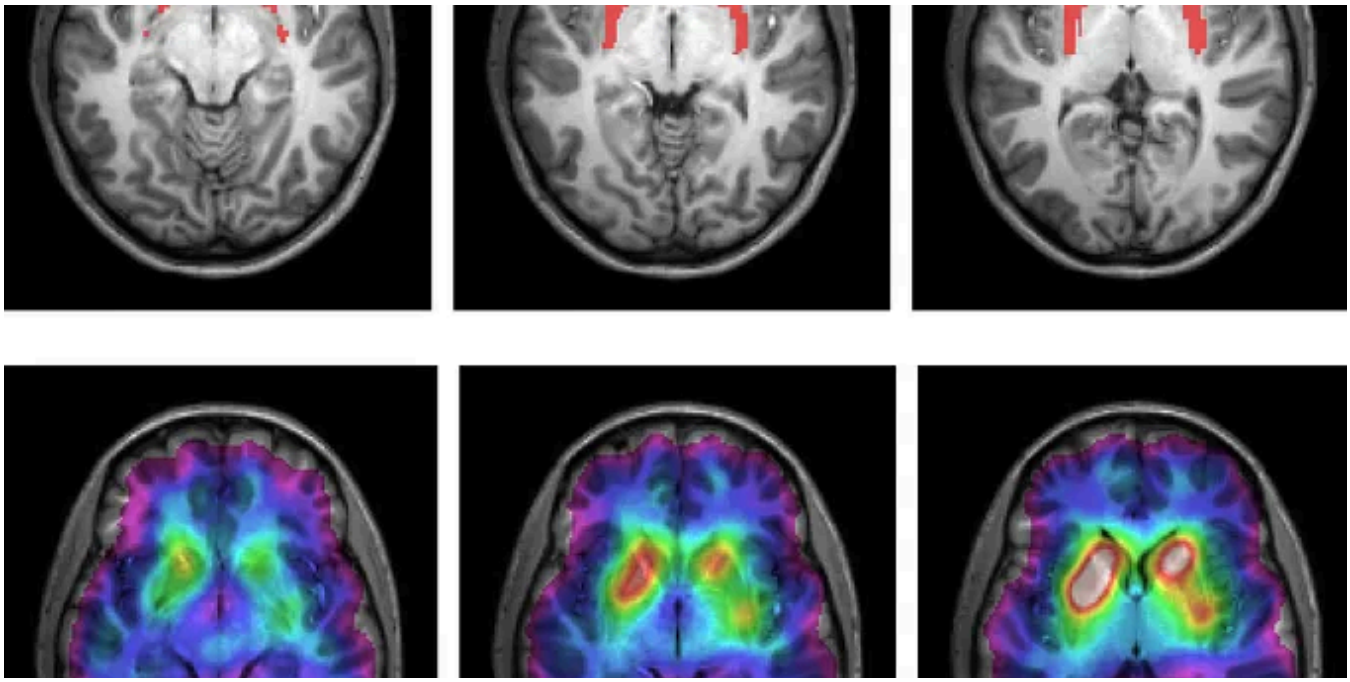

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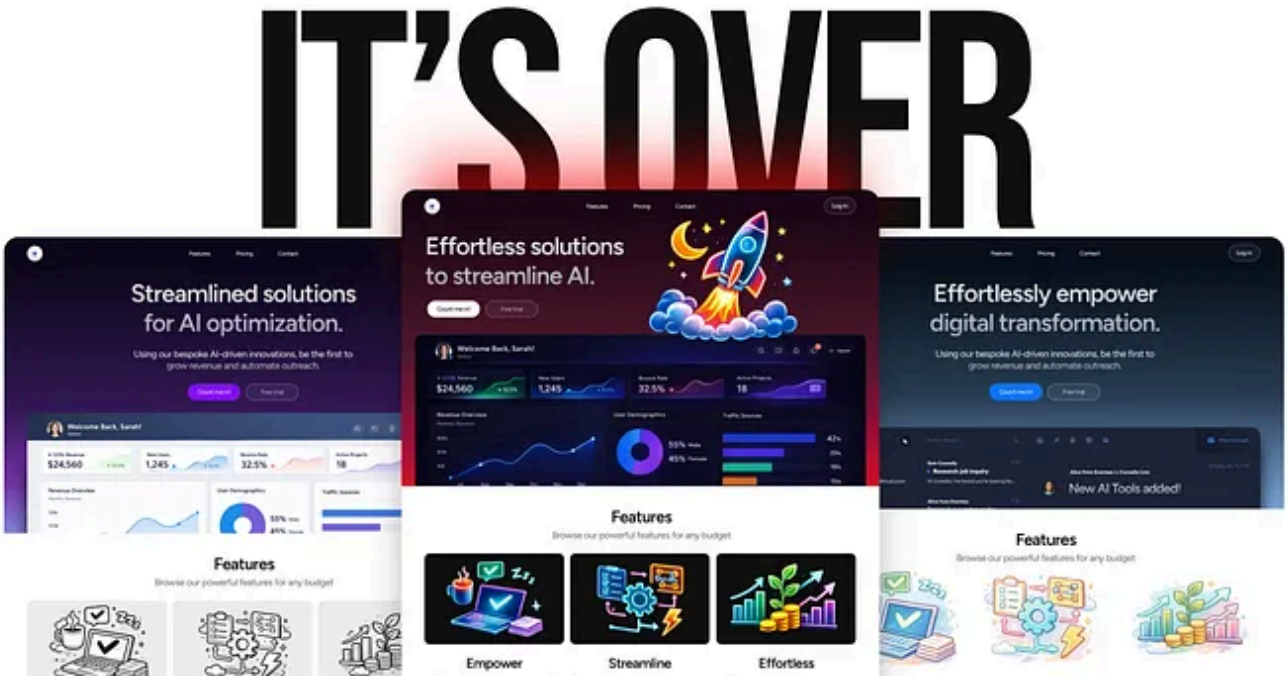

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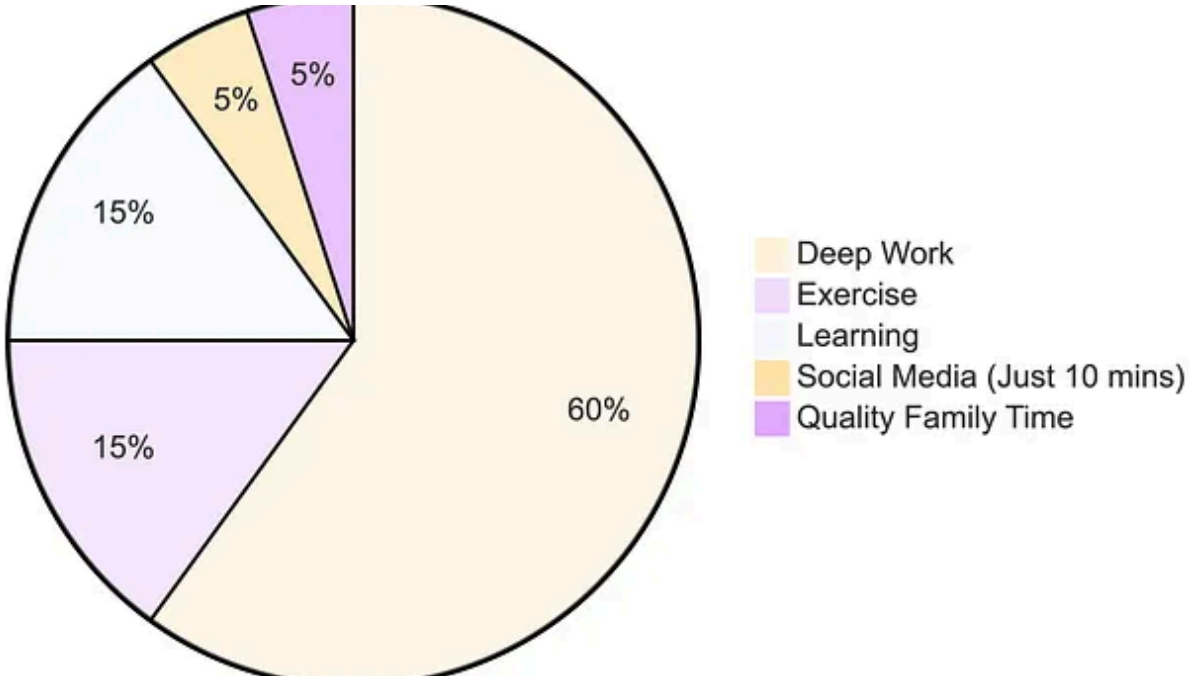

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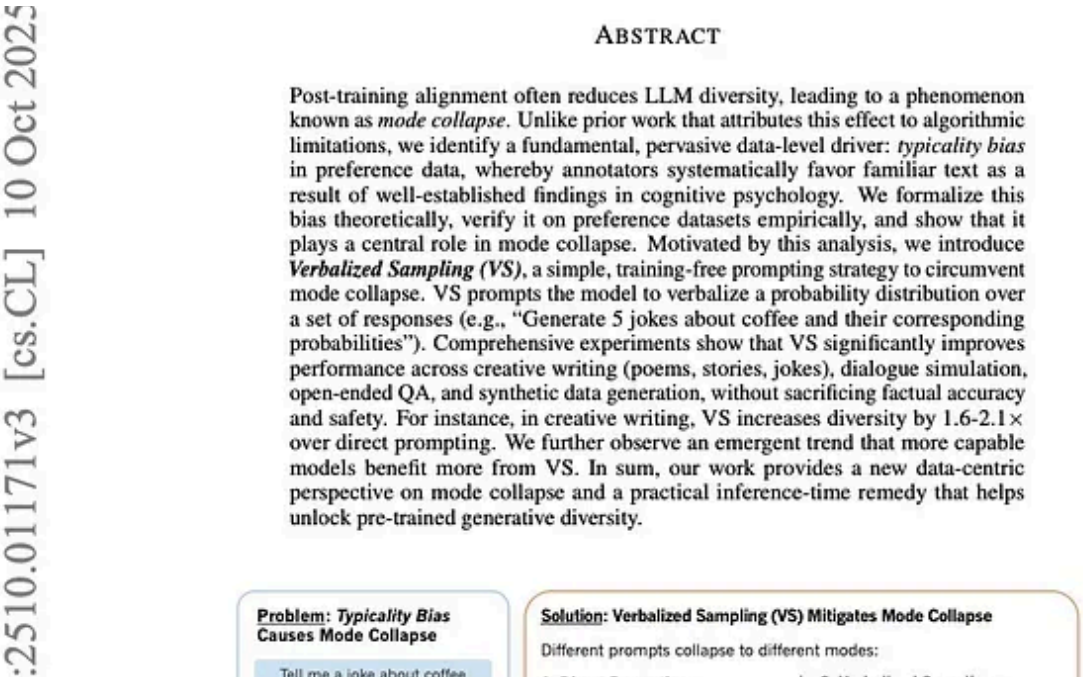

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ABSTRACT

Post-training alignment often reduces LLM diversity, leading to a phenomenon known as *mode collapse*. Unlike prior work that attributes this effect to algorithmic limitations, we identify a fundamental, pervasive data-level driver: *typicality bias* in preference data, whereby annotators systematically favor familiar text as a result of well-established findings in cognitive psychology. We formalize this bias theoretically, verify it on preference datasets empirically, and show that it plays a central role in mode collapse. Motivated by this analysis, we introduce ***Verbalized Sampling (VS)***, a simple, training-free prompting strategy to circumvent mode collapse. VS prompts the model to verbalize a probability distribution over a set of responses (e.g., "Generate 5 jokes about coffee and their corresponding probabilities"). Comprehensive experiments show that VS significantly improves performance across creative writing (poems, stories, jokes), dialogue simulation, open-ended QA, and synthetic data generation, without sacrificing factual accuracy and safety. For instance, in creative writing, VS increases diversity by 1.6-2.1× over direct prompting. We further observe an emergent trend that more capable models benefit more from VS. In sum, our work provides a new data-centric perspective on mode collapse and a practical inference-time remedy that helps unlock pre-trained generative diversity.

Problem: *Typicality Bias* Causes Mode Collapse

Solution: Verbalized Sampling (VS) Mitigates Mode Collapse

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